Anupam Paints

ANUPAM PAINTS 50+ PAINTING NATION FOR 50+ YEARS

Fire Seal Intumescent

Fire Retardant Coating

Product Description

Fire Seal is a water based intumescent fire retardant latex coating for interior and exterior (top-coating required) applications. When subjected to fire, it swells from 60 to 100 times its original thickness forming a stable and inert carboneous char which forms a physical barrier between the flame and the substrate. This char thermally insulates the substrate from heat and stops the availability of oxygen required for fire initiation. It provides fire protection to most metallic and non-metallic substrates. It provides effective heat shield and insulation to the substrates.

- **Introducing NanoShield Carbon Armor Technology for superior flame resistance.**

Technical Data

Type	Single Pack Cold Cured
Composition	Special Latex Polymer & Fire Retardant Chemicals
Color	As per requirement
Finish	Smooth & Matt
Volume Solids%	51 ± 2%
Wet Film Thickness (per coat)	1000 microns
Dry Film Thickness (per coat)	500 microns
Theoretical Coverage (no loss)	16-18 sqft/Liter in 2 coat basis
Dry Time @ 30° C & 50% R.H.	Touch dry: within 30 minutes To handle: within 90 minutes Hard dry: overnight
Application	By brush, roller & airless spray
Flash Point	> 93°C
Flame Resistance	120 mins
Salt Spray Test for 10 days.	No signs of blistering and corrosion observed
Toxicity	No toxicity

VOC Level: VOC content does not exceed 30 ppm & complies with applicable low-VOC requirements.

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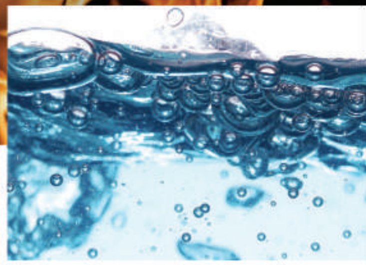

Surface Preparation

No special surface preparation is required. The surface should be clean and dry, free of dirt, oil, loose scale or paint and other foreign matter. On porous or flaky rusty surfaces, loose flakes and/or rusty scales must be removed by scraping and a proper surface suitable for the application of the coating restored. Coating on the primed surfaces must be sanded and/or primed and allowed to dry for 24 hours.

Application Procedure

Due to possible setting of pigment during shipping and storage, the product should be thoroughly mixed. Do not add water to the paint. If at all required, water to the tune of 5% maximum may be added. Apply a coat of Fire Seal liberally so that a dry film thickness of 250 microns is achieved and allow it to cure for 24 hours. Apply a second coat of 250 microns dry film thickness and allow the same to dry. The total DFT being 500 microns minimum.

Top Coating

Fire Seal can be top-coated with latex paints and two-component polyurethane paints for exterior applications.

Fire Test Results

Plywood panels of size 30x30x1 cm were painted with two coats of Fire Seal paint at dry film thickness of 250±10 microns per coat and the total dry film thickness being 500±20 microns. The test was carried out with the help of two feet tunnel equipment. A flame from a gas burner was applied to the coated plywood panels for 10 minutes. The temperature of the flame being 1400°C. The Panels were examined for height of intumescence formation. Further the char was removed to find out whether there was any damage to the plywood surface.

Fire temperature	1400°C
Flame speed length, cm	6.4
Char length, cm	4.2
After glow time, seconds	no after glow
Temperature on the reverse side of the panel	82°
Fuel gas temperature	79 °

Apart from the above test, intumescent property of Fire Seal coating was further proved by putting on flame the wooden huts coated with two coats of Fire Seal coating and the huts coated with conventional fire retardant paint. The huts were burnt by igniting 300 ml of alcohol kept inside the huts. After 15 minutes of test, the huts were examined for any damage to the structure. The hut coated with Fire Seal Coating was not damaged at all and there was no damage observed to the wooden surface of the hut. Whereas the hut coated with conventional fire retardant coating reduced to ashes within 15 minutes.

This product conforms to the criteria specified in IS 17044-2018, ASTM E119, and UL 263.

DISCLAIMER:
Information provided herein is based upon tests believed to be reliable. It does not guarantee the results to be obtained. Nor does it make any express and implied warranty or merchantability, or fitness for a particular purpose concerning the effects or results of such case. It does not release you from the obligation to test the products supplied by us to their suitability for the intended uses. The application, surface preparation and use of the products are beyond our control and, therefore, entirely your own responsibility.

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